HORN: Directional Outlier Propagation in Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. HORN tests whether attention-to-SSM and SSM-to-attention boundaries differ in outlier magnitude and quantization-noise sensitivity. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass activation-magnitude, quantization-noise propagation, and cross-model/control gates before GPU validation. H1a first screens one live hybrid model for directional activation statistics, H1 requires the same selected direction on at least two hybrid models, H2 tests matched FP4-equivalent noise propagation, and H3 requires the direction to survive hybrid-model repeats while weakening on pure-attention and pure-Mamba controls.

2 Mac Artifact Status

A deterministic synthetic H1a packet validates the directional-boundary real-schema contract. It emits 72 rows, selected max-abs ratio 4.044, non-boundary selected-direction control ratio 1.042, and permuted selected-direction control ratio 0.247, producing `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_HORN_H1A`. This is synthetic-only: it is not model evidence, not a quantization-noise result, and not GPU evidence. The metadata-only model eligibility packet identifies live hybrid targets and records architecture-map hashes for admissible packets, but no target weights are cached repo-locally, so HORN has not yet produced real boundary rows on the current Mac.

3 Next Evidence

The next admissible row is a real H1a boundary activation dump from a hybrid model. HORN remains a control branch unless real H1/H2/H3 gates show consistent directional asymmetry that pure-attention and pure-Mamba controls do not share. Pure-architecture controls are an H3 requirement, not a blocker for the first H1a boundary-activation packet.

4 Admissible Real Packet

The real H1a/H1 packet must use the shared config-derived architecture maps rather than substring-only module classification. Each row must report `model_id`, `prompt_id`, adjacent layer IDs, `direction`, `matched_boundary_direction`, `boundary_index`, normalization placement, `max_abs`, `rms`, `kurtosis`, and `control_type`. The required controls are boundary rows, non-boundary adjacent rows, and permuted-direction rows. A valid packet must cover both `attention->ssm` and `ssm->attention` in boundary rows and in matched control labels. Non-boundary controls may retain their true architecture direction, such as `ssm->ssm`, while `matched_boundary_direction` records the boundary direction they control for, and both matched directions must appear for every prompt; each permuted-direction row must match an observed boundary tuple with the same prompt ID, layer IDs, normalization positions, and measured metrics while flipping the actual `direction` label only. It cannot pass if the permuted controls preserve the selected high-magnitude direction; a faithful label flip may preserve unsigned asymmetry while moving the signal to the opposite label. Non-boundary controls must stay below the selected H1 threshold, not merely below the observed boundary ratio. It must also include at least 12 fixed prompt IDs unless the packet records an explicit resource limit. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. The packet must include `prompt_ids_hash`, `architecture_map_hash`, and a summary with prompt count, boundary directions, selected H1a ratio, selected H1a CI lower bound, support fraction, non-boundary control ratio, and permuted-direction ratio, then pass `experimental.shared.check_gate_packet --mode real --project horn`. The checker recomputes these H1a fields from raw rows and rejects stale or hand-filled summaries. Saved tensor packets should be converted with `experimental.shared.hybrid_trace_packet_builder` before validation.

Required control	Reviewer objection addressed
Explicit boundary map	Avoids substring artifacts.
Non-boundary adjacent pair	Separates boundary effects from depth effects.
Permuted direction label	Tests whether direction carries signal.
Matched normalization side	Controls RMSNorm/LayerNorm placement.
Pure-architecture control	Prevents generic outlier claims.

Table 1: Controls required before HORN can claim directional hybrid asymmetry.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/horn/paper/reviewer_pack.md`. It records that HORN is not camera-ready as a method or measurement paper until real H1–H3 evidence exists.